WATER RESOURCES, WELLS & PUMPS · NOTE WR04

Development of Irrigation in India — History, Plans and Cropping Pattern

Course topics: IDE5.5 L2, IDE8.16 L2

Ancient and medieval period

- Ancient Indian literature mentions wells, tanks and canals.
- Great tanks: the Viranarayana tank and the Gangaikonda-Cholapuram tank (Tamil Nadu) and the Anantaraja Sagara (Andhra Pradesh).

British period

- The British renovated and extended existing works. Between 1836 and 1866 they investigated, developed and completed four major diversion works: the Upper Ganga Canal, Upper Bari Doab Canal, Krishna Delta System and Godavari Delta System.
- The droughts and famines of 1876–78 forced further expansion.

End of the 19th century	Figure
Gross irrigated area	13.2 million ha (public works 7.5 million ha, private works 5.7 million ha)
Gross sown area	82.2 million ha — 16% irrigated
Share by source	Canals 45%, wells 35%, tanks 15%, others 5%

- The famines of 1897–98 and 1899–1900 led to the **First Irrigation Commission (1901)**. It recommended promoting private works and a 20-year plan to irrigate 2.6 million ha by public works.
- Irrigated area rose to 16 million ha by 1920–21.
- By 1945: wells irrigated 5.6 million ha and other private sources 4.4 million ha; the total irrigated area was 19.4 million ha, of which about 8 million ha now lie in Pakistan.

Five-Year Plans

- Partition (1947) sharply cut the irrigated area. In 1950–51 the net irrigated area was 20.85 million ha and the gross irrigated area 22.60 million ha.
- The Planning Commission gave irrigation priority — major projects such as Bhakra-Nangal, Damodar Valley and Hirakud.
- Gross irrigation potential grew from 22.6 million ha in 1950–51 to 90 million ha in 1996–97.

Minor irrigation

- Gained importance in the Grow More Food (GMF) Campaign (1943), with priority to tanks, open wells and tubewells.
- From 1951 to 1969, minor works brought 6.15 million ha under new irrigation; the total under minor sources reached 19.03 million ha.
- Growth was rapid in 1966–69 because of monsoon failures and the introduction of high-yielding varieties, and after 1970 because of better technology, improved drilling equipment and earth-

moving machinery.

Projects and sources by State

- States with many major and medium projects: Maharashtra, Madhya Pradesh, Gujarat, Rajasthan, Karnataka, Andhra Pradesh and Uttar Pradesh; Union Territories have very few.
- **Net area irrigated by source (2018–19):** wells, especially tubewells, dominate in most States; canals are significant in Uttar Pradesh, Punjab, Haryana and Rajasthan; tanks stand out in the south, especially Tamil Nadu; the eastern and north-eastern States rely more on other sources and rain-linked systems.

Cropping pattern (1992–93) — a historical snapshot

Item	Share of total cropped area (185 million ha)
Food grains	70% (cereals 57%, pulses 13%)
Rice / wheat	23% / 13%
Oilseeds	14%
Net area sown ÷ total cropped area	77%

- From 1991–92 to 1992–93: rice area fell by 7000 ha; wheat rose by 11,000 ha (≈ 5%); gram rose by 13%.
- Rice held 23% from 1950–51 to 1992–93; wheat held 13% from 1980–81 to 1992–93 (it was 8–11% from 1950–51 to 1970–71).
- Sorghum (jowar), pearl millet (bajra), barley and coarse cereals declined; gram fell from 6% (1950–51) to 3% (1992–93).
- Sugarcane rose from 1.3% to 2.2%; fruits and vegetables from 1.7% to 3.7%; oilseeds from 8% to 14%; cotton stayed near 4%.
- Area sown more than once rose from 10% (1950–51) to 23% (1992–93), mainly because irrigation allowed a second crop.

These cropping figures describe the early 1990s; use the latest Land Use Statistics of the Ministry of Agriculture for current values.

Sources: ICAR eCourse 'Minor Irrigation and Command Area Development' — Dr K. Shanmuga Sundaram (TNAU), reviewed by Dr P. Balakrishnan (UAS Karnataka). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. The First Irrigation Commission was appointed in

(a) 1867 (b) 1901
(c) 1947 (d) 1972

Q2. Gross irrigated area in 1950–51 was about

(a) 13.2 million ha (b) 22.6 million ha
(c) 56 million ha (d) 90 million ha

Q3. The Grow More Food campaign that boosted minor irrigation began in

(a) 1901 (b) 1943
(c) 1966 (d) 1978

Q4. At the end of the 19th century, the largest share of irrigated area came from

(a) wells
(b) tanks
(c) canals
(d) others

Q5. Which was NOT among the four diversion works completed in 1836–1866?

(a) Upper Ganga Canal
(b) Upper Bari Doab Canal
(c) Indira Gandhi Canal
(d) Godavari Delta System

Answers

Q1 — (b) 1901

Q2 — (b) 22.6 million ha

Q3 — (b) 1943

Q4 — (c) canals — *Canals 45%, wells 35%, tanks 15%.*

Q5 — (c) Indira Gandhi Canal
